INTERNSHIP REPORT

ON

"CAD DESIGNING — SKETCHING, PART MODELLING,

ASSEMBLY & ENGINEERING DRAWING"

Submitted in partial fulfilment of the requirements of the

CAD Designing Internship Program

Submited by

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CERTIFICATE

STUDENT DECLARATION

I, Sahil Yadav, a student of B.Tech (Mechanical Engineering), 3rd Year, Rajkiya Engineering College, Banda, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, hereby declare that the work presented in this internship report titled "CAD Designing — Sketching, Part Modelling, Assembly & Engineering Drawing" is an original record of the work carried out by me during my internship at Nexonix Reflexo Pvt. Ltd. under the CAD Designing Internship Program 2026.

I further declare that this report, or any part of it, has not been submitted earlier for the award of any degree, diploma, or certificate to this or any other institute, and that all the assignments, sketches, part models, assemblies, and engineering drawings included in this report were completed by me during the course of the internship, under the guidance of my company mentor and faculty mentor.

Date:

Place:

SAHIL YADAV

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Nexonix Reflexo Pvt. Ltd. for providing me the opportunity to undertake this CAD Designing Internship and for the valuable exposure to industry-relevant design tools and workflows.

I am deeply thankful to my company mentor, Mr. Anmol Verma, for his constant guidance, patient explanations, and constructive feedback throughout the internship, which greatly enhanced my understanding of CAD designing, sketching fundamentals, part modelling, assembly design, and engineering drawing practices.

I would also like to thank my faculty mentor and the Department of Mechanical Engineering, Rajkiya Engineering College, Banda, for their continuous support and encouragement in pursuing this internship alongside my academic curriculum.

I am grateful to my institute, Rajkiya Engineering College, Banda, and Dr. A.P.J. Abdul Kalam Technical University, Lucknow, for fostering an environment that encourages practical, industry-oriented learning.

Finally, I extend my thanks to my teammates and fellow interns for their cooperation, healthy discussions, and shared learning throughout the duration of this internship.

SAHIL YADAV

ABSTRACT

This report presents a detailed account of the CAD Designing Internship undertaken at Nexonix Reflexo Pvt. Ltd., a four-week, industry-oriented training program focused on building practical competence in Computer-Aided Design using SolidWorks, AutoCAD, Onshape, Tinkercad and FreeCAD. The internship was structured around progressive, lecture-linked assignments that moved from fundamental 2D sketching to advanced part modelling, and finally to multi-part assembly design with production-ready engineering drawings.

In the first week, sketching fundamentals were reinforced through three basic 2D profiles — a rectangle, a ring, and an L-shaped bracket — followed by an independent Mounting Bracket sketch challenge that tested command over sketch entities, geometric constraints, and dimensioning needed to achieve a fully defined sketch. The second week extended this foundation into advanced part modelling, where a Symmetrical Mounting Bracket was designed using Mirror and Flex features to create a 90° bend with a reinforcing rib, and a thinwalled Phone Stand was modelled using circular pattern, shell, draft angle, and fillet features to produce a manufacturing-ready component. The third week consolidated assembly design, engineering drawing, and surface modelling concepts through a capstone project — a multi-part Belt Support Assembly — where individual components were modelled and constrained using mates into a fully defined assembly, and documented through a complete orthographic engineering drawing with GD&T annotations. Throughout the internship, emphasis was placed on maintaining clean, error-free feature trees, applying correct geometric and dimensional constraints, and exporting industry-standard file formats (.SLDPRT, .SLDASM, .STL, .STEP, .SLDDRW) alongside dimensioned screenshots for evaluation. The internship also strengthened soft skills such as time management, selfdirected learning, and technical documentation.

This report documents the objectives, weekly activities, project details, tools used, results achieved, skills gained, and overall learning outcomes of the internship, and concludes with a reflection on the practical value of CAD design training in preparing for a career in mechanical design engineering.

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CHAPTER 1: INTERNSHIP OBJECTIVES

The primary objective of this internship was to build a strong, practical foundation in Computer-Aided Design (CAD) through structured, hands-on assignments covering sketching, part modelling, assembly design, and engineering drawing. The following sections outline the specific objectives achieved during the course of the internship.

Technical Skills

During the internship, I developed proficiency in 2D sketching (entities, constraints, dimensioning), 3D part modelling (extrude, pattern, mirror, shell, fillet, draft, flex, rib), assembly modelling using mates, and the generation of production-ready engineering drawings with orthographic views and basic GD&T, using SolidWorks as the primary tool along with Tinkercad, AutoCAD, Onshape and FreeCAD. Specific technical objectives achieved include:

- Creating fully defined 2D sketches using geometric and dimensional constraints.
- Building parametric 3D parts using extrude, pattern, mirror, shell, fillet, draft, flex, and rib features.
- Assembling multiple parts using mates and verifying degrees of freedom.
- Generating orthographic engineering drawings with basic GD&T annotations.
- Exporting industry-standard file formats such as .SLDPRT, .SLDASM, .STL, .STEP, and .SLDDRW.

Practic al Learning

Each assignment was structured around a real, manufacturable component rather than an abstract exercise, which helped me understand how design decisions — such as draft angles, fillets, and wall thickness — connect to manufacturability. I also learned the importance of maintaining a clean, error-free FeatureManager tree and verifying that sketches and assemblies were fully defined before export. Key practical takeaways included:

- Sequencing features correctly so that later operations (mirror, shell, draft) do not break earlier ones.
- Anchoring the first sketch entity to the origin so assemblies and drawings stay stable and predictable.
- Treating screenshots and exported files as deliverables in themselves, not an afterthought — matching how real design teams document their work.

Tea m Collaboration

Although each assignment was individual work, I took part in discussions with fellow interns to compare modelling approaches, troubleshoot feature errors, and review each other's screenshots before submission, which improved my ability to communicate technical work clearly.

Problem Solving

Several tasks required independent problem-solving — for instance, achieving a fully constrained sketch for the Mounting Bracket without step-by-step guidance, or resolving rebuild errors after applying the Shell and Draft features on the Phone Stand. These situations strengthened my debugging approach: isolating the failing feature, reviewing the sketch/feature dependency order, and applying corrections systematically.

CHAPTER 2: WEEKLY INTERNSHIP ACTIVITIES

This chapter summarises the learning topics, assignments, skills gained, and outcomes for each week of the internship.

Week 1 (Day 1–7): Sketching Fundamentals — "Sketch It Right"

Learning Topics: Introduction to CAD Designing and sketching fundamentals — sketch entities, constraints, and dimensioning.

Assignment: Assignment 1 – "Sketch It Right: 2D Profile & Bracket Challenge", consisting of Part A (three basic 2D profiles: a 60×40 mm rectangle, a Ø50/Ø20 mm ring, and an 80×20 + 20×50 mm L-shape) and Part B (an independent Mounting Bracket sketch with a base plate, an upright arm, two Ø10 mm holes, and perpendicular/coincident constraints).

Skills Learned: Ruler and Align tools, precise dimension entry, concentric and coincident placement, applying perpendicular constraints, and verifying fully defined sketch status. **Outcome:** Successfully completed and submitted all three basic profiles and the independent bracket sketch, achieving a fully defined sketch as required by the assignment.

Week 2 (Day 8–14): Advanced Part Modelling — "Build It Better"

Learning Topics: Part modelling basics and advanced part modelling — Linear Pattern, Mirror, Shell, and Draft Angle features.

Assignment: Assignment 2 – "Build It Better: Advanced Part Modelling Challenge", consisting of Task 1 (a Symmetrical Mounting Bracket built using the Mirror feature for bolt holes and a Flex feature for a 90° bend, with a mirrored support rib) and Task 2 (a thinwalled Phone Stand built on a filleted equilateral-triangle base, with a circularly patterned slot, a drafted back support, a shell feature, and edge fillets).

Skills Learned: Mirror and circular pattern features, the Flex feature for bending plate-like bodies, the Rib feature with draft, the Shell feature for wall-thickness reduction, and Fillet/Draft for manufacturability.

Outcome: Exported both parts as .SLDPRT/.STL files with clean, error-free FeatureManager trees showing the Shell, Fillet, and Extrude features, as required by the submission checklist.

Week 3 (Day 15–21): Assembly, Drawing & Surface Design — "Build It Complete"

Learning Topics: Assembly design, engineering drawings, and surface modelling.

Assignment: Assignment 3 – "Build It Complete: Assembly, Drawing & Surface Design Challenge", consisting of Task 1 (a multi-part Belt Support Assembly built by modelling individual components and constraining them using mates, with one fully defined and one movable configuration) and Task 2 (a complete, production-ready engineering drawing of the assembly, with orthographic views and GD&T basics).

Skills Learned: Applying mates and joints, achieving zero degrees of freedom in an assembly, generating orthographic, section, and detail views, and applying basic GD&T callouts on an engineering drawing.

Outcome: Completed a fully constrained assembly and its corresponding engineering drawing, exported in .SLDASM/.STEP and .SLDDRW/PDF formats respectively.

Week 4 (Day 22–28): Consolidation, Documentation & Reporting

Learning Topics: Reviewing all completed assignments, organising exported files and screenshots, and compiling internship documentation.

Activities: Organised all part files, assembly files, STL/STEP exports, and dimensioned screenshots into a public GitHub repository; posted process screenshots on LinkedIn as required; and compiled this internship report summarising the objectives, weekly activities, project work, tools used, and outcomes of the internship.

Skills Learned: Version-control basics (GitHub), professional documentation, and technical report writing.

Outcome: Successfully consolidated four weeks of CAD design work into a structured, submission-ready portfolio and this internship report.

CHAPTER 3: PROJECT INTRODUCTION — BELT SUPPORT ASSEMBLY

Among the assignments completed during the internship, the Belt Support Assembly from Assignment 3 (Week 3) is presented here as the primary project, since it integrates part modelling, assembly design, mating, and engineering drawing into a single, productionrepresentative deliverable.

Objectives

The objective of this project was to design a multi-part Belt Support assembly — comprising a pulley/wheel, a bushing, a shaft, and bracket plates — model each part individually, assemble them using appropriate mates so that the assembly is fully constrained (zero degrees of freedom) in one configuration while remaining movable in a second configuration, and finally generate a complete, production-ready engineering drawing of the assembly with orthographic views and basic GD&T annotations.

Scope

The scope of this project was limited to the CAD design and documentation of the Belt Support assembly — individual part modelling, assembly-level mating, and 2D engineering drawing generation — using SolidWorks (with Onshape, AutoCAD, and FreeCAD as acceptable alternative tools). Manufacturing, material selection, and stress analysis of the assembly were outside the scope of this assignment and were not undertaken.

Applications

A belt/pulley support assembly of this type is commonly used in belt-drive power transmission systems, conveyor systems, and general industrial machinery, where it supports a rotating shaft or pulley and transfers load safely to a mounting structure. Understanding the design and documentation of such an assembly is directly applicable to real-world mechanical design roles involving power transmission components, material handling systems, and machine design.

CHAPTER 4: HARDWARE / SOFTWARE USED

Hardware Table

	Specification / Purpose
Laptop / Desktop Computer	[Add processor / RAM details] — used for running CAD software
Mouse	Used for precise sketch creation and 3D model navigation
Internet Connection	Used for accessing cloud-based tools (Onshape) and submitting assignments via GitHub / LinkedIn

Software Table

	Purpose
SolidWorks	Primary tool for sketching, part modelling, assembly design, and engineering drawing
AutoCAD	Used for 2D drafting and alternative sketching exercises
Tinkercad	Used for basic 2D profile sketching and the Mounting Bracket challenge (Assignment 1)
Onshape	Alternative cloud-based CAD tool used for modelling and assembly tasks
FreeCAD	Alternative open-source CAD tool used where applicable
GitHub	Used to host and submit exported part/assembly files (.SLDPRT, .SLDASM, .STL, .STEP, .SLDDRW)
Microsoft Word	Used to prepare this internship report

Programming Languages

No programming languages were used during this internship, as all assignments were focused on CAD sketching, part modelling, assembly design, and engineering drawing rather than software development or automation scripting.

Libraries

Not applicable. Standard SolidWorks Toolbox components and default material libraries were used where required, rather than external programming libraries.

Tools (CAD Features / Commands Used)

- Sketch tools: Line, Rectangle, Circle, Ruler, Align, Smart Dimension
- Constraints: Coincident, Perpendicular, Concentric, Fixed
- Part modelling features: Extrude Boss/Cut, Linear Pattern, Circular Pattern, Mirror, Shell, Fillet/Round, Draft, Flex, Rib
- Assembly tools: Mates (Coincident, Concentric, Perpendicular) and Degrees-ofFreedom check
- Drawing tools: Orthographic view projection, Section view, Detail view, and GD&T annotation tools

CHAPTER 5: RESULTS AND DISCUSSION

Testing

Before exporting each part, sketch, and assembly, the FeatureManager Design Tree was checked for rebuild errors and warnings (red/yellow flags), and each sketch was verified to be Fully Defined (all sketch entities shown in black rather than blue). For the Belt Support assembly, one configuration was verified to have zero degrees of freedom (fully mated), while a second configuration was kept movable to demonstrate correct mate behaviour.

Output

The final outputs included: three basic 2D profiles and an independent Mounting Bracket sketch (Assignment 1); a Symmetrical Mounting Bracket and a thin-walled Phone Stand with Shell, Fillet, and Draft features (Assignment 2); and a fully constrained Belt Support assembly along with its production-ready engineering drawing (Assignment 3).

Graphs

As this internship was focused on CAD design rather than data analysis, conventional graphs were not applicable. Instead, dimensional verification tables and constraint/mate status checks (Fully Defined / Under Defined / Over Defined, and Degrees of Freedom = 0) served the same verification role — confirming that each model met its specified dimensions and constraints before submission.

Challenges

[Add any specific challenges you personally faced — for example, initial difficulty achieving a fully defined sketch without over-constraining it, or resolving a rebuild error after applying the Shell feature on the Phone Stand.]

In general, common challenges in this type of work include the following, each of which was resolved by reviewing the sketch/feature dependency order, adjusting constraint sequencing, and cross-checking with the assignment guidelines:

- Over- or under-constraining a sketch, leading to a status other than Fully Defined.
- Feature rebuild errors caused by the order in which operations (extrude, mirror, shell, draft) were applied.
- Correctly identifying and selecting reference planes for the Mirror and Flex features.
- Achieving zero degrees of freedom in the assembly while keeping a second, movable configuration available.
- Fitting all required orthographic views, section views, and GD&T callouts clearly onto a single drawing sheet.

Future Improvements

Future work could extend the Belt Support assembly with motion simulation to verify pulley rotation and belt clearance, basic stress/FEA analysis on the bracket components, and photorealistic rendering for presentation purposes. Additional practice with sheet-metal design and top-down assembly modelling would also further strengthen design skills.

CHAPTER 6: SKILLS LEARNED

Technical Skills

2D sketching and constraint-based parametric modelling; part modelling features including pattern, mirror, shell, fillet, draft, flex, and rib; assembly modelling with mates; engineering drawing generation with orthographic views and basic GD&T; and file management/export in industry-standard formats (.SLDPRT, .SLDASM, .STL, .STEP, .SLDDRW). Summarised by assignment:

- Assignment 1: Sketch entities, geometric constraints, precise dimensioning, fully defined sketch status.
- Assignment 2: Mirror, Circular Pattern, Flex, Rib, Shell, Fillet, and Draft features.
- Assignment 3: Assembly mates, degrees-of-freedom verification, orthographic views, and basic GD&T.

Soft Skills

Time management (meeting weekly assignment deadlines), self-directed learning (completing the independent Mounting Bracket challenge without step-by-step guidance), attention to detail (achieving fully defined sketches and error-free feature trees), and organisation (maintaining a structured GitHub submission repository). These soft skills were built gradually, week over week, alongside the technical skills, rather than being taught separately:

- Planning each week's work around the assignment due date and submission checklist.
- Working through the independent Mounting Bracket challenge (Assignment 1, Part B) without step-by-step instructions.
- Reviewing my own FeatureManager tree critically before every submission to catch errors early.

Communication

Improved ability to document and present technical work clearly — through dimensioned screenshots, structured file naming, and concise written summaries of each assignment for LinkedIn posts and this report.

Leadership

[Add any specific instance, e.g., helping a fellow intern debug a feature error or explaining a modelling approach to the group.]

While this was primarily individual work, I supported peer learning by discussing modelling approaches and troubleshooting steps with fellow interns.

Industrial Exposure

The structure of the internship — lecture-linked assignments, weekly deadlines, submission checklists, and grading criteria — mirrored real industry design workflows, giving practical exposure to how CAD design tasks are scoped, executed, documented, and reviewed in a professional setting.

CHAPTER 7: CONCLUSION

Summary

Over four weeks, this internship progressed systematically from 2D sketching fundamentals to advanced part modelling and, finally, to multi-part assembly design with engineering drawings. Each assignment built directly on the skills developed in the previous one, culminating in the Belt Support Assembly project, which integrated part modelling, assembly mating, and engineering drawing into a single, production-representative deliverable.

LearningO utcomes

By the end of the internship, I was able to independently create fully defined 2D sketches, model manufacturable parts using advanced features such as shell, draft, mirror, and pattern, constrain multi-part assemblies using mates, and generate production-ready engineering drawings — skills that are directly relevant to a career in mechanical design engineering. The internship also reinforced the importance of clean modelling practice, systematic problemsolving, and clear technical documentation.

CHAPTER 8: REFERENCES

IEEE Format

[1] SOLIDWORKS Corporation, SOLIDWORKS User Guide, Dassault Systèmes, 2024.

[2] American Society of Mechanical Engineers, ASME Y14.5-2018: Dimensioning and Tolerancing, ASME, New York, 2018.

APA Format

Dassault Systèmes. (2022). SOLIDWORKS user guide. Dassault Systèmes SolidWorks Corporation.

American Society of Mechanical Engineers. (2018). ASME Y14.5-2018: Dimensioning and tolerancing. ASME.

Books

- Shigley, J. E., & Mischke, C. R. — Mechanical Engineering Design, McGraw-Hill Education.
- Bhatt, N. D. — Machine Drawing, Charotar Publishing House.

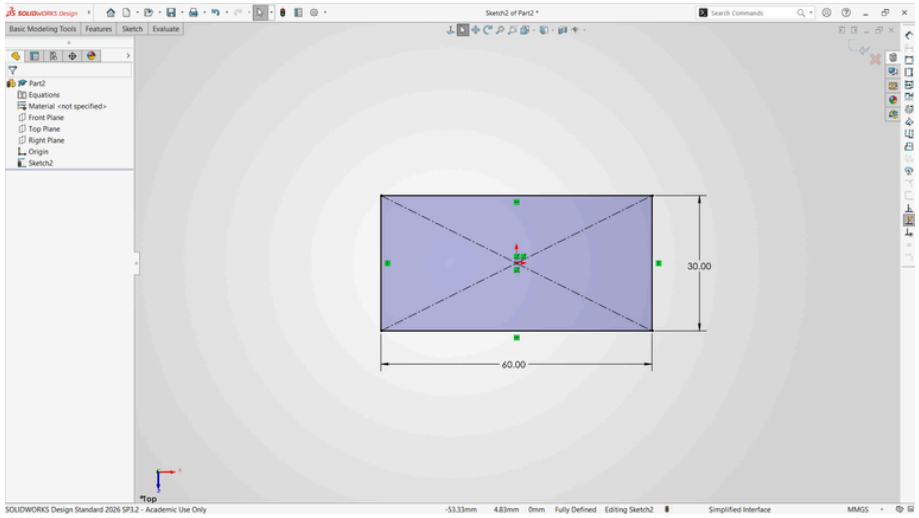
Websites

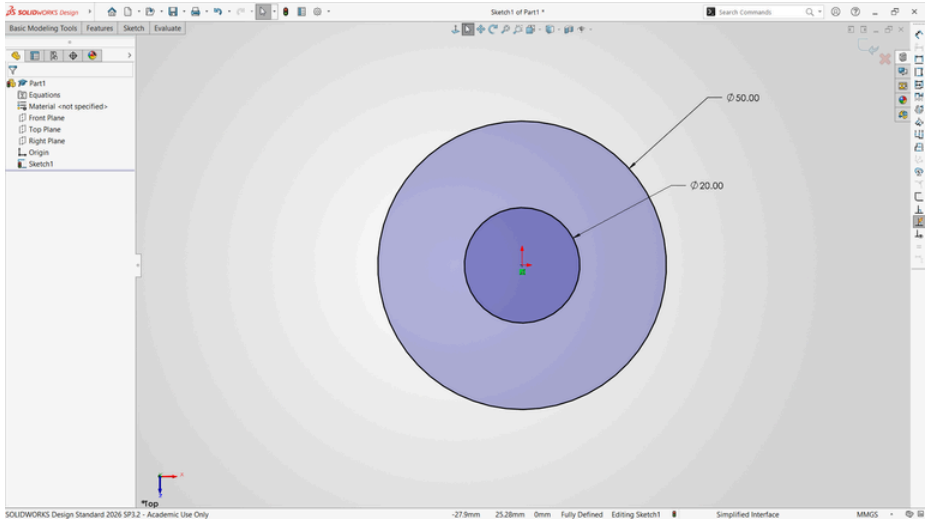
- SOLIDWORKS Official Help & Documentation — https://help.solidworks.com
- GrabCAD Community — https://grabcad.com

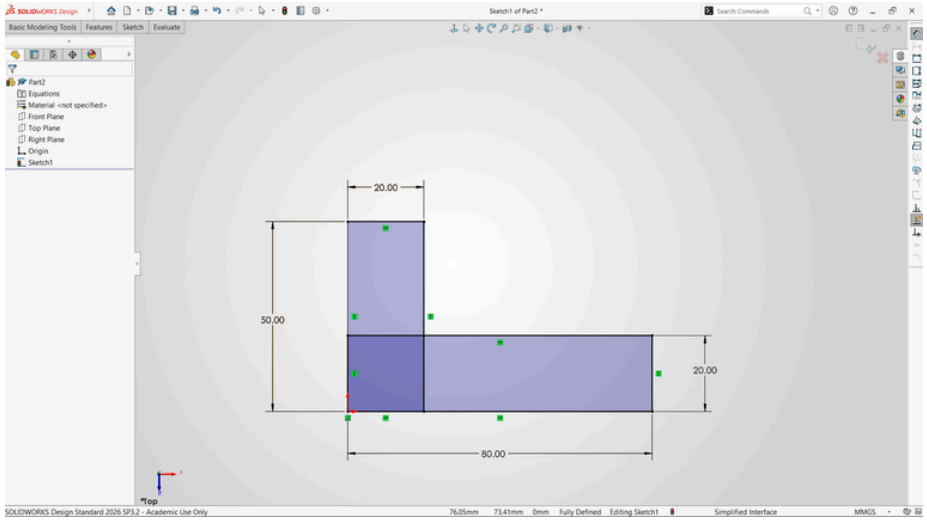
Research Papers

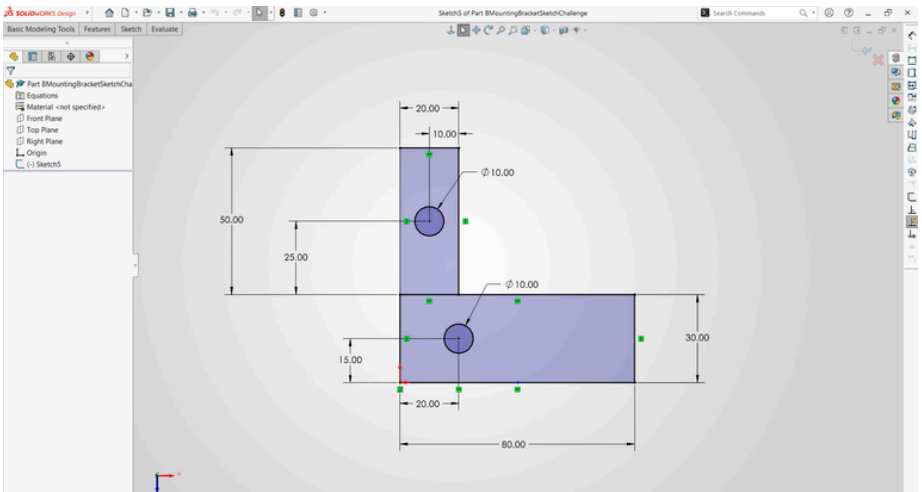
[Add any research papers referred to during the internship, if applicable.]

WEEK-1

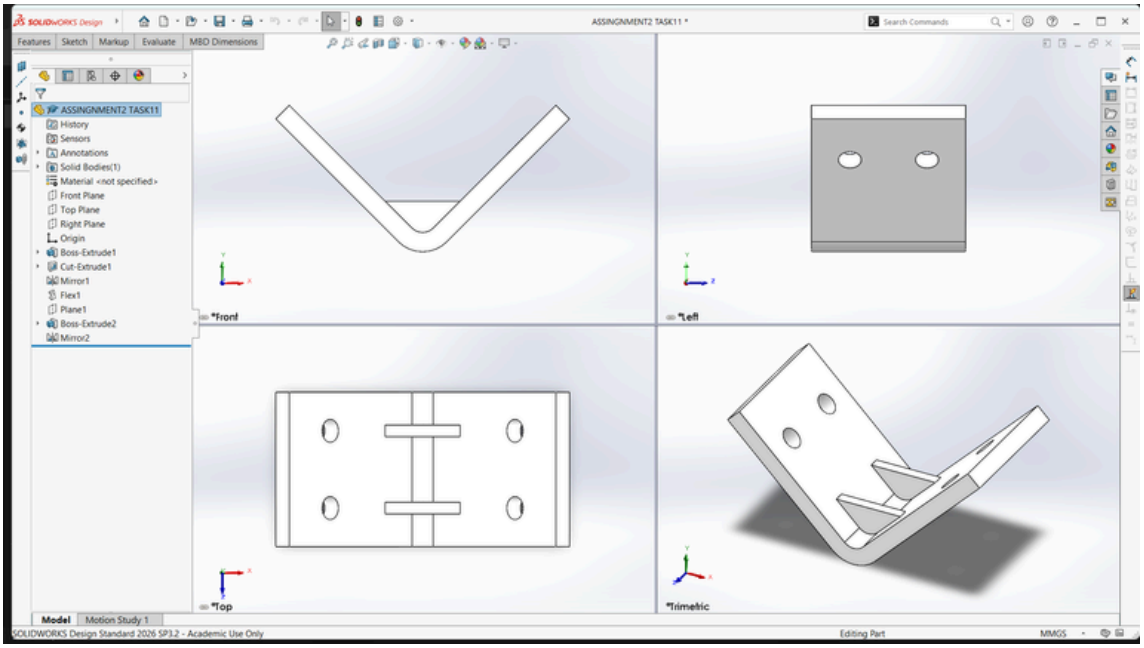

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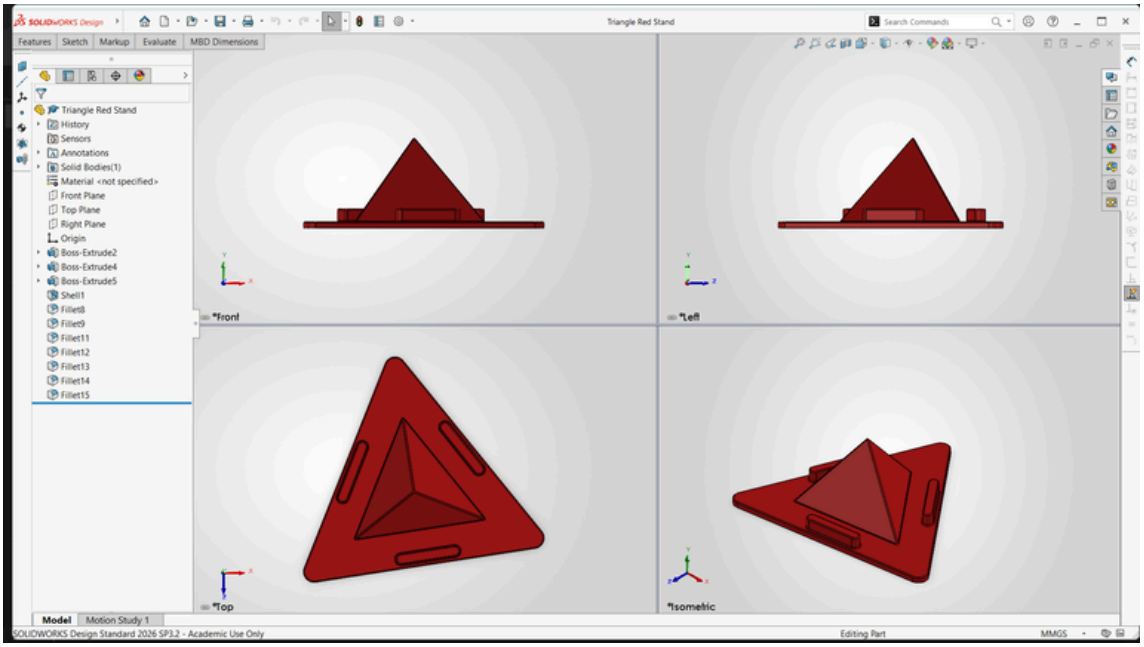

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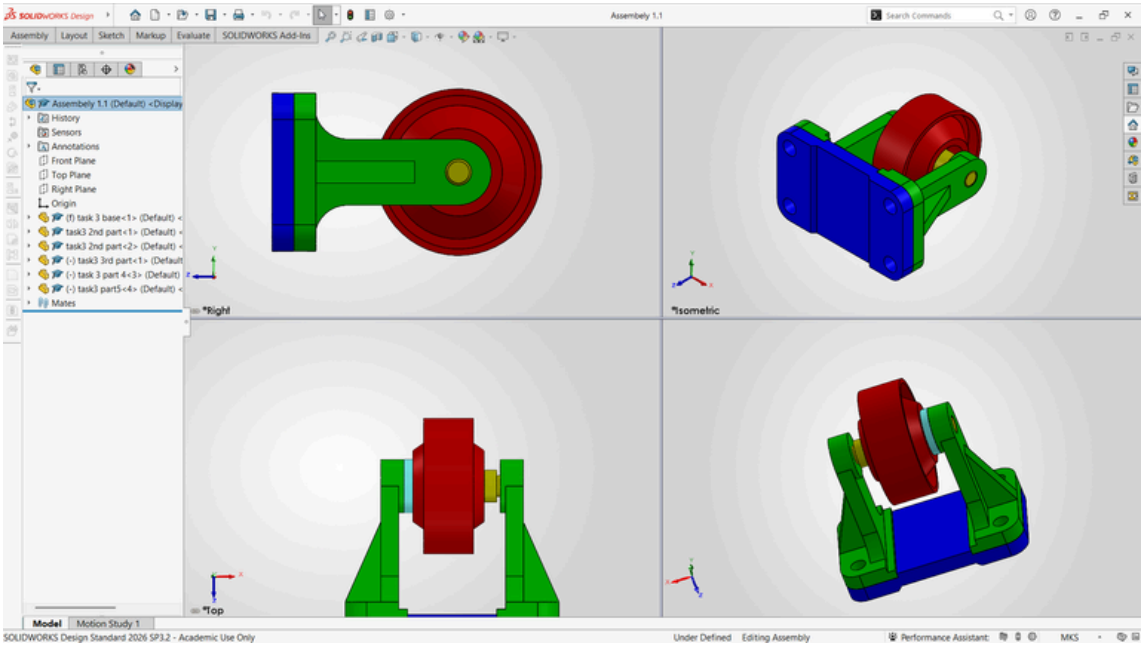

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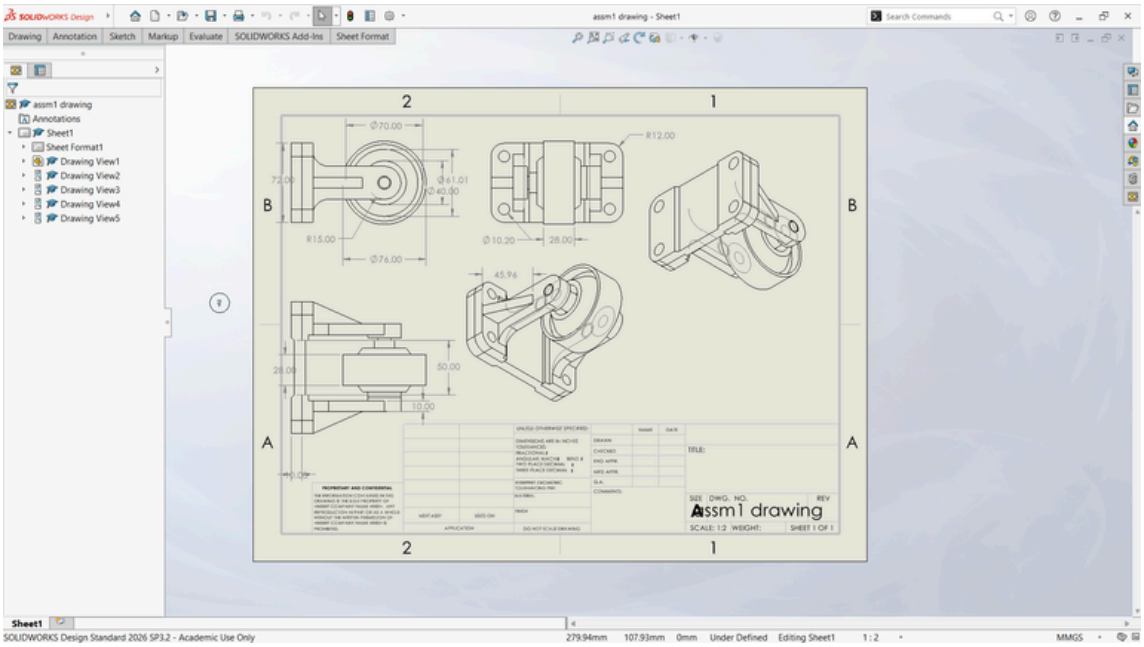
WEEK-2


ASSIGNMENT2 TASK11 *
Features
Sketch
Markup
Evaluate
MBD Dimensions
ASSIGNMENT2 TASK11
History
Sensors
Annotations
Solid Bodies(1)
Material <not specified>
Front Plane
Top Plane
Right Plane
Origin
Boss-Extrude1
Cut-Extrude1
Mirror1
Flex1
Plane1
Boss-Extrude2
Mirror2
*Front
*Left
*Top
*Trimetric
Model
Motion Study 1
SOLIDWORKS Design Standard 2026 SP3.2 - Academic Use Only
Editing Part
MMGS


Triangle Red Stand
Features
Sketch
Markup
Evaluate
MBD Dimensions
Triangle Red Stand
History
Sensors
Annotations
Solid Bodies(1)
Material <not specified>
Front Plane
Top Plane
Right Plane
Origin
Boss-Extrude2
Boss-Extrude4
Boss-Extrude5
Shell1
Fillet8
Fillet9
Fillet11
Fillet12
Fillet13
Fillet14
Fillet15
*Front
*Left
*Top
*Isometric
Model
Motion Study 1
SOLIDWORKS Design Standard 2026 SP3.2 - Academic Use Only
Editing Part
MMGS

WEEK-3


Assembly 1.1
*Right
*Isometric
*Top
Model
Motion Study 1
SOLIDWORKS Design Standard 2026 SP3.2 - Academic Use Only
Under Defined
Editing Assembly


assm1 drawing - Sheet1
assm1 drawing
Sheet1
Assm1 drawing
SOLIDWORKS Design Standard 2026 SP3.2 - Academic Use Only